DERIVED PUSHFORWARD OF A PERFECT COMPLEX ALONG A FLAT PROPER MORPHISM OF NOETHERIAN SCHEMES IS PERFECT

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Perfectness criterion for derived pushforwards. lemma-perfect-direct-image (same chapter, NOT carded, declared external). Three hypotheses; (1) and (2) are immediate here, (3) is local on X.
- Strictly perfect representatives. structural. Affine locally, E is represented by a strictly perfect complex of O_X-modules.

Source reference. Original lemma 08EV: sources/stacks-perfect/pinned-original-source.tex, source lines 6990–6990. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a Noetherian scheme. Let $f : X \to S$ be a flat proper morphism of schemes. Let $E \in D(\mathcal{O}_X)$ be perfect. Then Rf_*E is a perfect object of $D(\mathcal{O}_S)$.*

Proof. We claim that Lemma 08EV applies. Conditions (1) and (2) are immediate. Condition (3) is local on X. Thus we may assume X and S affine and E represented by a strictly perfect complex of $\mathcal{O}_X$-modules. Since $\mathcal{O}_X$ is flat as a sheaf of $f^{-1}\mathcal{O}_S$-modules we find that condition (3) is satisfied. □

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